

RMO– 2024

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let $n > 1$ be an integer. Call a rearrangement $a_1, a_2, a_3, \dots, a_n$ of $1, 2, 3, \dots, n$ nice if for every $k = 2, 3, 4, \dots, n$ we have $a_1 + a_2 + a_3 + \dots + a_k$ is not divisible by k .
(a) If $n > 1$ is odd, prove that there is no nice arrangement of $1, 2, 3, \dots, n$.
(b) If n is even, find a nice arrangement of $1, 2, 3, \dots, n$.
2. For a positive integer n , $R(n)$ be the sum of remainders when n is divided by $1, 2, 3, \dots, n$. For example $R(4) = 0 + 0 + 1 + 0 = 1$, $R(7) = 0 + 1 + 1 + 3 + 2 + 1 + 0 = 8$. Find all positive integer n such that $R(n) = n - 1$.
3. Let ABC be an acute triangle with $AB = AC$. Let D be the point on BC such that AD is perpendicular to BC . Let O, H, G be the circumcenter, orthocenter and centroid of triangle ABC respectively. Suppose that $2 \cdot OD = 3 \cdot HD$. Prove that G lies on the incircle of triangle ABC .
4. Let a_1, a_2, a_3, a_4 be real numbers such that $a_1^2 + a_2^2 + a_3^2 + a_4^2 = 1$. Show that there exist i, j with $1 \leq i < j \leq 4$, such that $(a_i - a_j)^2 \leq \frac{1}{5}$.
5. Let $ABCD$ be a cyclic quadrilateral such that AB is parallel to CD . Let O be the circumcenter of $ABCD$, and L be the point on AD such that OL is perpendicular to AD prove that $OB(AB + CD) = OL(AC + BD)$.
6. Let $n \geq 2$ be a positive integer. Call a sequence $a_1, a_2, a_3, \dots, a_k$ of integers an n -chain if $1 = a_1 < a_2 < a_3 < \dots < a_k = n$, and a_i divides a_{i+1} for all i , $1 \leq i \leq k - 1$. Let $f(n)$ be the number of n -chains where $n \geq 2$. For example $f(4) = 2$ corresponding to the 4-chains $\{1, 4\}$, $\{1, 2, 4\}$. Prove that $f(2^m \times 3) = 2^{m-1}(m + 2)$ for every positive integer m .

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